

Telemetry goes in. A person signs the work order.

PM Triage turns condition-monitoring telemetry into an evidence-backed maintenance case. **Deterministic rules detect** the abnormal reading. **Physics rules plus one narrow trained model name the fault — or abstain.** A language model retrieves precedent, explains the evidence and drafts the work order. A **named planner must approve** before an idempotent adapter writes anything to the maintenance system of record. No AI step can move a machine, set a priority, or authorize work. That boundary is structural, not a policy note.

VERIFIED

BACKEND TESTS

105

all passing

FRONTEND

Build ✓

Next.js production build

DEPLOYED

Live

Vercel · Render · Supabase

DEFAULT RUN COST

\$0.00

mock mode even with a key

FOUR OWNERS — AND NO OWNER CAN QUIETLY TAKE ANOTHER'S JOB

■ DETERMINISTIC CODE

Is this reading abnormal? How urgent is it?

detector.py · classifier.py
priority.py · calibration.py

■ TRAINED MODEL

One job only: suction vs discharge restriction on the SKAB pump.

ml_classifier.py
Extra Trees + IsolationForest

■ LANGUAGE MODEL

Precedent, explanation, recommended actions, work-order prose.

agent/triage.py · agent/llm.py
three read-only tools

■ NAMED HUMAN

Does any work actually happen? Every case, every time.

POST /cases/{id}/decision
token + reviewer name

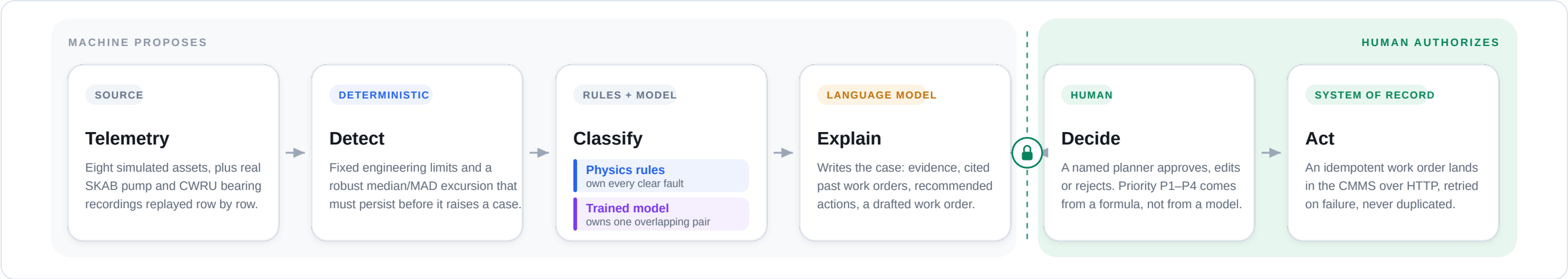
WHAT IS REAL

Application code and workflow	Production code running on Vercel and Render
Database	Supabase Postgres via SQLAlchemy; SQLite locally and in tests
SKAB / CWRU telemetry	Recorded laboratory data, curated into eight replay episodes
CMMS interface	Real HTTP contract, schema mapping, retries, idempotency

WHAT IS NOT — SAY THIS BEFORE YOU ARE ASKED

CMMS system	A mock FastAPI app mounted in the same deployment, sharing the database
Fleet telemetry	Eight simulated assets: Gaussian baselines plus deterministic fault ramps
CWRU transition	Real healthy and faulty steady states concatenated — not a natural run-to-failure
Maintenance history	Seeded realistic demo records, not a customer CMMS export

One abnormal reading, left to right



Everything left of the dashed line is a proposal. A work order is created in exactly one place, after exactly one state transition, by a request carrying a bearer token and a reviewer name — no autonomous path from telemetry to action.

WHO DECIDES WHAT

DECISION	OWNER
Is this reading abnormal?	Deterministic code — fixed limits, sustained robust median/MAD
What kind of fault is it?	Physics rules first; the trained model only for the overlapping pair
Is the evidence strong enough to answer?	Calibration on precedent and signature; below 0.45 it abstains
How is it explained, what is proposed?	The language model — mock or live, identical tool protocol
How urgent is it?	A published formula. The agent may move it one notch, never below P1
Does any work actually happen?	A named human, on every case, including high-confidence ones
Where does approved work land?	The CMMS, over HTTP, idempotently

WHY THE BOUNDARIES SIT THERE

- **Detection is deterministic** because it must be cheap, continuous and readable by a technician. The same reading always gives the same result.
- **The trained model is deliberately narrow.** A novelty score and an exact sensor-roster check make it abstain outside its job rather than answer confidently in a domain it never saw.
- **The language model never decides**, so a fluent wrong answer costs an explanation, not a maintenance action. And **confidence is grounded, not self-reported** — a model's own certainty tracks fluency, not correctness.

Priority is arithmetic, not prose.
score = criticality + 2×severity + min(recurrence,3) + 4
if safety
P1 if safety or ≥13 · P2 ≥10 · P3 ≥7 · else P4

Every box is a real file or a real service

BROWSER

Next.js App Router on Vercel — / · /cases · /cases/[id] · /machines/[id] · /cmms · /audit · /eval. lib/api.ts is the only network layer: one base URL, a bearer token on every state-changing call.

HTTPS · JSON

↓ BEARER TOKEN ON EVERY WRITE ↓

BACKEND

FastAPI on Render — one ASGI app, one async event loop. Telemetry keeps ticking while a serial worker performs triage; one physical machine event produces exactly one case.

release-pinned

INGEST

simulator.py

eight assets, one reading every 3 s; detection runs on every reading

replay.py

real SKAB pump and CWRU bearing episodes, replayed row by row

POST /simulate/inject

faults are manual in production — SPONTANEOUS_FAULT_PROB=0

retention.py

raw telemetry pruned after 24 h; cases and audit events are records

DETECT + CLASSIFY

detector.py

fixed limits + robust median/MAD, sustained; one case per event

classifier.py

signature rules over cross-signal trends — owns the clear faults

ml_classifier.py

Extra Trees, trained on one job: suction vs discharge restriction

novelty + roster guard

IsolationForest and an exact feature roster; unknown input abstains

AGENT

agent/triage.py

bounded tool loop; a provider failure falls back to the free policy

agent/tools.py

machine info · recent telemetry · history search · recurrence count

agent/llm.py

mock policy or OpenRouter — identical tool protocol

agent/calibration.py

grounds confidence in precedent and signature; below 0.45 it abstains

DECIDE + WRITE BACK

priority.py

P1–P4 from criticality, severity, recurrence, safety. Agent may ±1

POST /cases/{id}/decision

the only path to approval; needs a token and a named reviewer

agent/cmms_adapter.py

httpx + Idempotency-Key + retry, only after an approval

audit.py

append-only: system, agent, human:<name>, with a timestamp

app.mount("/cmms") → cmms/service.py — a separate ASGI service with its own work-order store. The triage backend reaches it **over HTTP with an idempotency key and a retry, never by importing it**. Point CMMS_BASE_URL at SAP or Maximo and nothing above this line changes. P1–P4 becomes CMMS priority 1–4; temperature maps to OHE, pressure and flow to LOO, vibration to VIB, cavitation to CAV; the case id becomes the idempotency key, so a retry returns the same order instead of creating a duplicate.

SUPABASE POSTGRES · SCHEMA PM_TRIAGE

machines · telemetry · anomalies · triage_cases · llm_calls · maintenance_logs · audit_events.

SQLite locally and in tests — one domain model either way.

OPENROUTER · DEEPSEEK/DEEPSEEK-V4-FLASH · LIVE MODE ONLY

Off by default: LLM_MODE=mock even when a key exists. 12 calls/day · \$0.25/day · 700 output tokens per call. Every request is costed into llm_calls; over cap, the case finishes free and deterministic and says so in its trace.

GITHUB ACTIONS

keep-warm.yml — a health ping every 10 minutes so the free Render instance answers a demo. It never runs triage. eval.yml — the offline evaluation suite.

Did it notice?

Right, over everything

How often it spoke

Right when it spoke

FIVE SKAB PUMP + THREE CWRU BEARING EPISODES

Six categories, and what each one is allowed to claim

CATEGORY	WEIGHT	EVIDENCE IN THIS ARTIFACT	DO NOT OVERCLAIM
Solution architecture & workflow design	30%	One owner per decision; the full telemetry-to-CMMS state machine; a fallback at every failure point.	A focused working slice, not a complete plant platform.
Software engineering & SDLC	25%	Typed FastAPI and Next.js contracts, 105 backend tests, a production build, live deployment, retention.	No formal SRE maturity or exhaustive browser E2E testing.
Systems integration & enterprise architecture	20%	REST APIs, Supabase Postgres, an HTTP anti-corruption adapter, schema translation, retries, idempotency.	The mock CMMS is mounted in the same deployment and shares the database.
Domain-embedded business architecture	15%	Criticality, engineering limits, signal features, precedent, P1–P4 priority, downtime exposure, a human planner.	Thresholds, history and economics are demo inputs, not validated policy.
Agentic AI / LLM engineering	7%	A bounded tool loop over three read-only tools, structured output, mock/live parity, an eval suite, provider fallback.	The LLM owns no detection, no authorization, and cannot overrule a classifier verdict.
Governance, observability & control	3%	OOD rejection, abstention, mandatory approval, an append-only audit trail, call and dollar caps.	Demo authentication is a signed crew-password session, not SSO or RBAC.

LIMITATIONS — VOLUNTEER THESE BEFORE YOU ARE CHALLENGED

- **Eight laboratory episodes are development evidence**, not customer-site validation — and the frozen restriction test is only three physical recordings.
- **The trained layer solves one pair**. Everything else is a rule, or an abstention. **CWRU episodes are constructed transitions**, with redistribution terms still to confirm.
- **Process memory is not shared state**. The replay cursor and triage queue live in one process; multi-replica needs a durable broker and a crash-recovery scanner.
- **Schema setup uses create_all**, not Alembic migrations. **The frontend polls**; it does not consume an event stream.

THE SENTENCE TO CLOSE ON

Rules detect. Rules plus narrow ML classify. The OOD gate can abstain. The LLM explains and retrieves. A named human authorizes. An idempotent adapter integrates the approved decision. That division is the production design — not a limitation of the prototype.

Next: customer-owned natural fault-onset recordings, especially discharge and cavitation; an external validation set that tests the thresholds without changing them; then monitor acceptance, selective accuracy and drift in production.